

x Description

For the Kaggle submission, please make sure to submit as a team with team name in the format "VU-DM-2026-Group-x", replacing x with the group number from Canvas, e.g. VU-DM-2026-Group-132. If these do not match we can't take your score into account when grading.

Start of assignment: April 20 2026, 00:00

Deadline Kaggle: May 17 2026, 23:55

Deadline Report Canvas May 24 2026, 23:59

This document describes Assignment 2 of the Data Mining Techniques course at the VU. Please make sure you read it thoroughly and carefully. This is a group task (3 members), and please make sure all team members contribute to the work as expected. There will be three things to be submitted: 1) A report about the results; 2) A file to be uploaded on the VU DMT Kaggle competition, and 3) A process report. The entire description of this assignment can be found on Canvas under the Page Assignment 2.

Motivation

By now should have a fair idea about techniques we can use, and should also have some practical experience with mining datasets. In this second assignment, you will gain more experience, explore various techniques (and whether they work in this situation), and hopefully learn a lot. The topic of this assignment is positioned in the area of recommender systems. More specifically, your task is to predict what hotel a user is most likely to book. This could greatly help companies such as Expedia (from which the dataset actually originates) to organize the search results for a user in the most suitable way.

Detailed Task Description

To make things easier, we will use a DM process model to describe your task in a bit more detail, similar to what you have seen in assignment 1, and during the lectures.

TASK 1: BUSINESS UNDERSTANDING

Your task is to predict what hotel properties that result from a search of a user, the user is most likely to click on. Of course, more people have worked on such predictions. Can you find some other people that have tried to make such predictions (e.g. from the Kaggle competition)? And what have they used as most prominent predictors? Have other people that participate in the competition

mentioned anything about their approaches? Please spend a couple of paragraphs on this topic in a 'Related Work' section in your report.

TASK 2: DATA UNDERSTANDING

In this task, you will do some exploratory data analysis (EDA). Explore the dataset, count, summarize, plot things, and report findings that are useful for creating predictions. Remember that EDA is not necessarily done once at the start of the project. It is expected that you do some EDA, build some features, train some models, then some idea comes up, do some more EDA, modify your features, train another model on these new features, and so on.

TASK 3: DATA PREPARATION

You'll certainly need to work on the dataset, to create, modify or add new features. For instance, you might want to compare the different properties that resulted from the search instead of learning from them one by one. There are certain attributes with a large amount of missing values, do they still provide useful information? And how will you handle a missing value if this shows to be the case? Finally, in order to test your approach (since you do not know the answers for the test set) you will need to split up your data to test your approach yourself before you generate your answers on the test set.

Of course, you are also allowed to use external data sources if you find ones that are useful. In case you get inspired by previous work on this competition, please make sure you properly cite the sources you base your approach on.

TASK 4: MODELING AND EVALUATION

Naturally, once you prepare the dataset, you should be able to build models. You have great freedom to select the techniques you feel are most appropriate and should try at least two different techniques. At least use one technique that has been discussed during the lecture on Recommender Systems or a variant thereof. The choice of that techniques or alternatives you try might be influenced by how we would like to measure your predictions at the end (described later in this document). To test how your model performs compared to other groups, you can upload your answers for the test set on the in class Kaggle website. Note that the score on Kaggle is only for part of the test set, the score on the rest of the test set will only be disclosed on the final lecture and will form part of your final grade.

TASK 5: DEPLOYMENT

While we do not go into real deployment, we do want you to:

- Apply one of the ethical AI techniques discussed in class to detect and mitigate possible biases in your model. You may choose any type of bias (e.g., variable performance across

user groups like business travelers vs. families or underrepresentation of certain hotels). In your report, describe:

1. How you detected the bias (include relevant fairness metrics)
2. Which bias mitigation technique you applied (pre-processing or post-processing)
3. How effective the mitigation was based on your evaluation results

Acknowledgements

This assignment is inspired in the original [Kaggle's Expedia Hotel Recommendations](#) Competition.

Deliverables and Formats

We have covered the process above, let us see what we expect you to deliver.

1. Predictions

You'll need to submit a single file, which ranks the properties belonging to a user search on the likeliness that the property will be booked. Here, you should start with listing the property most likely to be booked. An example of part of such a file is shown below.

```
srch_id , prop_id
2, 7771
2, 26540
2, 25579
2, 7374
2, 131173
2, 37331
2, 27090
2, 12938
2, 78858
2, 30434
2, 91899
2, 3105
2, 6399
3, 130729
3, 103937
3, 556881
```

2. Scientific Report

The assignment is not only about winning, but also about quality of the process and understanding of what you did. Therefore, we would like you to write a report, which should contain the following:

1. What you did (you might want to follow the process model, and describe the steps you took. If you tried a number of things but only some worked, please mention those that did not work as well, and discuss why they might not have worked).
2. What you learned (either inside the main part of the report, or separately in a paragraph of two, please describe what skills and knowledge you have gained from this assignment, what were the main difficulties, expected and unexpected outcomes of your experiments, etc.
3. Please format the report according to the Incs guidelines (exactly the way you used previously when doing assignment 1).

The report should be maximum 14 pages long. Again, it is 14 pages with everything included except for the references.

3. Process Report

As the assignment is done in a group, we would like to get insight into what each individual group member contributed to the eventual result. Therefore we ask you to compose a process report of at most 2 pages which addresses:

1. A schedule of when you performed what task (e.g. on April 28 we explored the dataset and looked for suitable approaches for the task at hand).
2. Who contributed to what task (e.g. Mark was responsible for transforming the dataset into a suitable format for SPSS Modeler whereas Berend was working on the report).
3. A critical reflection of the overall cooperation within the team.

Submission

The submission consists of two parts:

1. the prediction file (on Kaggle)
with deadline 17/05/2026, 23:55
2. the written reports (via Canvas)
with deadline 24/05/2026, 23:59

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Closing Event

The presentation of your final assignment will be done during the closing event. This will take place in IN-0B60 on May 20 (from 11:00-12:45). Here, we will present the final outcome of the competition (given the predictions you handed in the weekend before) and hand out the cup and the

fame and glory to the lucky winners. Six groups will be asked to present, those groups that ended up in the top 3 and three additional random groups will be asked to present their work.

Evaluation

Here's how you will get rewarded for your work:

1. Winning The Competition

The winner will be rewarded with the fame and glory of winning the 2026 VU Data Mining Techniques cup. Your accuracy score will be determined as follows (cf. Kaggle):

The evaluation metric for this competition is Normalized Discounted Cumulative Gain (NDCG)@5 calculated per query and averaged over all queries with the values weighted by the `log_2` function. See https://en.wikipedia.org/wiki/Discounted_cumulative_gain#Normalized_DCG for more details. Hotels for each user query are assigned relevance grades as follows:

```
5 - The user purchased a room at this hotel
1 - The user clicked through to see more information on this hotel
0 - The user neither clicked on this hotel nor purchased a room at this hotel
```

Submissions for each user query should recommend hotels in order from the highest grade (most likely to purchase a hotel room) to the lowest grade (least likely to purchase a hotel room or click on the hotel). We know that the correct values for the test set are available online, of course, you are not allowed to use this. If we suspect that you used those values we will ask you for your code and check whether your results are reproducible using the training set as a basis for generating your predictive model. You should upload your prediction on this website.

2. Getting a High Grade

You can get 100 marks for this assignment, so half of your final grade depends on this. 80% of this mark can be achieved by submitting a nice and thorough report, 20% will come from where you end up in the competition. Regarding the competition-based marks, we will see how good your submissions are, and then we'll decide if you'll get the marks based on accuracy value, or rank in the competition.

Dataset Description

The dataset originates from a Kaggle competition, is split into a training and a test set (training_set_VU_DM.csv and test_set_VU_DM.csv respectively, each containing approximately 5 million records). Essentially, the dataset contains information about a search query of a user for a hotel, the hotel properties that resulted and for the training set, whether the user clicked on the hotel and booked it.

1. File descriptions

- training_set_VU_DM.csv - the training set
- test_set_VU_DM.csv - the test set
- submission_sample.csv - a sample submission file in the correct format

2. Data fields

For a very nice overview on the data fields of the task, please refer to <https://www.kaggle.com/c/expedia-personalized-sort/data>.

Each line in the dataset represents a combination of a search query by a user with one specific hotel property that was shown as part of the results. Of course, a list of hotels is presented to the user (and hence, there are multiple rules describing a single search). Lines that belong to the same user/search are identified by the same search id.

The link between the data fields and the Expedia site are illustrated in <https://www.kaggle.com/c/expedia-personalized-sort/data> as well.

2.1. Training set

- position - (Integer) - Hotel position on Expedia's search results page. This is only provided for the training data, but not the test data.
- click_bool - (Boolean) - 1 if the user clicked on the property, 0 if not.
- booking_bool - (Boolean) - 1 if the user booked the property, 0 if not.
- gross_booking_usd - (Float) - Total value of the transaction. This can differ from the price_usd due to taxes, fees, conventions on multiple day bookings and purchase of a room type other than the one shown in the search.